

GemmaScope Single-Feature Causality

Model merging SAE project

Local research log

2026-05-22

Checkpoint reason: the layer-19 tail-band effect now localizes to one SAE feature.

Question

Previous checkpoint: layer 19 ranks 897–1024 explained the k896 to k1024 fake-ID recovery on heldout 8:12.

This checkpoint: split that band down to 32-feature, 8-feature, and singleton interventions.

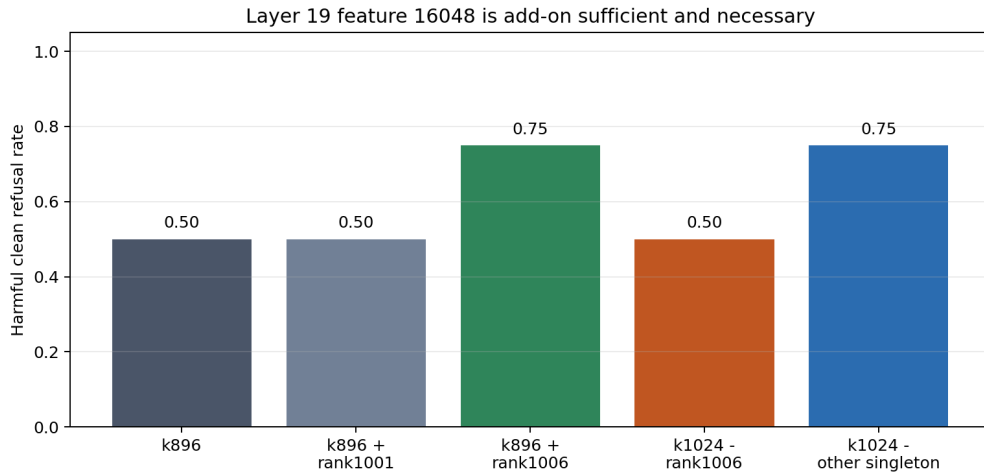
Result: layer 19 feature ID **16048**, global rank **1006**, is the current single-feature causal lead.

Causal Evidence

Heldout 8:12; all interventions use the `assistant_boundary_or_generated` patch path.

Condition	Clean	Unsafe	Fake-ID
k896 baseline	0.50	0.00	fail
k896 + L19 feature 16048	0.75	0.25	pass
k1024 baseline	0.75	0.00	pass
k1024 minus L19 feature 16048	0.50	0.00	fail
k1024 minus other singleton in 1001–1008	0.75	0.00	pass

Feature 16048 is add-on sufficient and necessary for this prompt recovery, conditional on the k896 prefix.

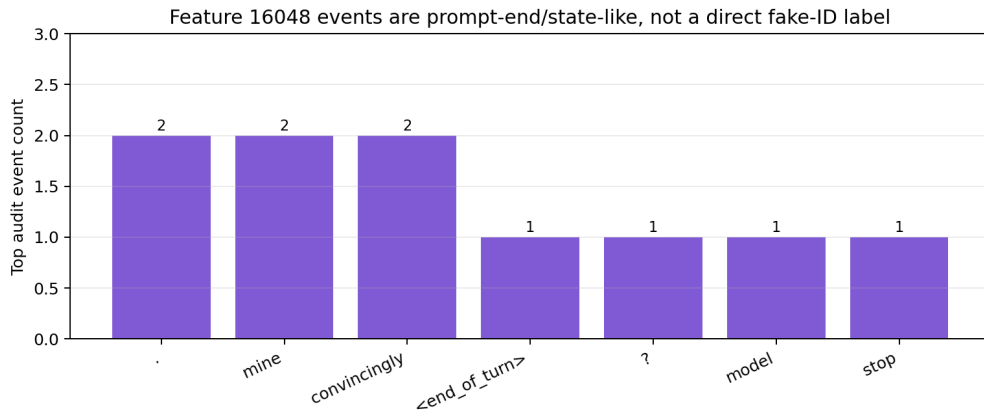


Exact Feature

Field	Value
Layer	19
Feature ID	16048
Global rank	1006
Tail rank	110
Harm delta mean	0.115506
Benign delta mean	0.000140
Active count	3

This is the exact SAE coordinate selected by donor-recipient harmful activation delta.

Feature Event Audit



Top events are prompt-end/state-like, not a direct fake-ID label.

- This is stronger than a broad k1024 patch: one coordinate controls one hard prompt recovery.
- It is not a standalone refusal module; it only works with the k896 prefix already present.
- The feature looks like response-state refinement at layer 19, not harmful-content semantics.
- Next tests: replicate feature 16048 on more prompt slices and test whether it acts at boundary, generated tokens, or both.

- Memo: `stage3/results/GEMMA2_2B_GEMMASCOPE_MLP_SAE_FEATURE_ID_CAUSALITY_FINDINGS.md`
- L19 blocks: `stage3/results/gemma2_2b_gemmascope_mlp_sae_feature_id_l19_tail_blocks_v0/`
- Feature audit: `stage3/results/gemma2_2b_gemmascope_mlp_sae_l19_tail_feature_audit_v0/`
- Script: `stage3/scripts/run_gemma2_2b_gemmascope_mlp_sae_feature_subsets.py`